



Dipartimento di Elettronica, Informazione e Bioingegneria
Politecnico Di Milano

Lab Experience

Final project presentation

Name of the project: Radar-Aided 6G Beam Prediction

Presenters:

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Academic Year:

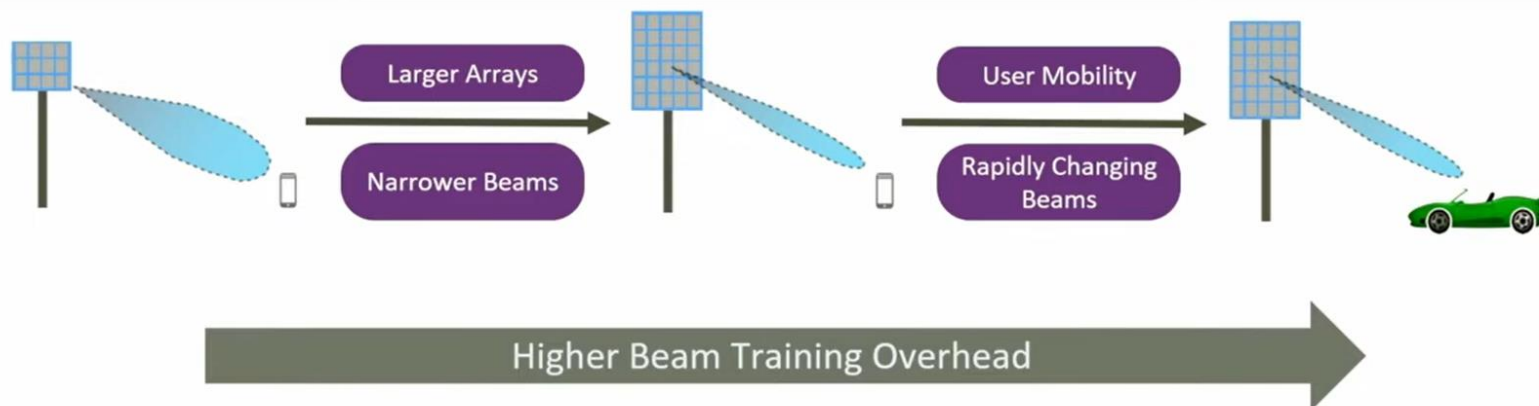
2023/2024



- Introduction
 - Context, problem, solution proposed
 - Lab XP and project objectives
- Theory recap
 - Adopted system, radar measurement, and communication model
 - SUMO
 - Radar signal preprocessing
- Implementation
 - Framework of the solution
 - Deep Learning: Inputs and models
- Results
 - Model comparison
- Conclusions



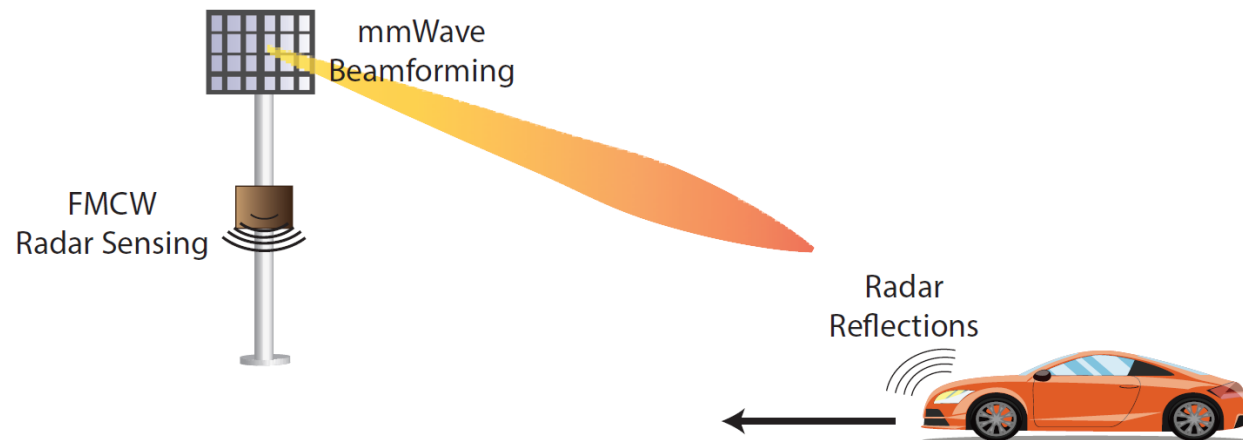
- **Context:** Millimeter-wave (mmWave) and sub-terahertz (sub-THz) communications are key pillars for current and future wireless communication networks.
 - Large available BW enable satisfaction of increasing data rate demands.
 - Beneficial for apps such as autonomous driving, edge computing, mixed-reality, etc.
 - Require deploying large antenna arrays at the transmitters and/or receivers and using narrow beams to guarantee sufficient receiver power.
- **Problem:**
 - Higher path loss at higher frequencies
 - User mobility makes it more challenging
 - High beam training overhead (search time to find/align the best beams)



■ Possible solution: Sensing-Aided Communications

- Awareness about their locations and the surrounding environment (geometry of the buildings, moving scatterers, etc.) could potentially help the beam selection process.
- The geometry of the environment determines the channel information and beams
- Sensing modalities can provide info about the geometry of the environment
- Sensing can be utilized for beam alignment problem

■ Radar-aided beam prediction at the infrastructure



Why radar?

- Low cost
- High-freq measurement
- Velocity measurements (preferable for highly mobile scenarios)
- No privacy concern



Team Objectives:

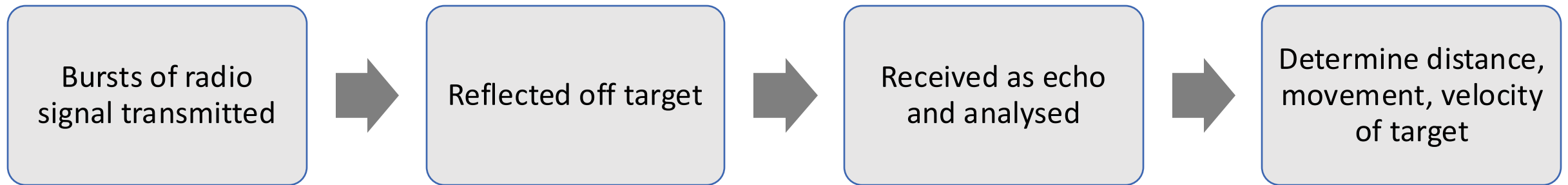
- Scenario modelling for radar measurement and communication between the stationary unit and mobile unit
- Visualization of the acquired measurements
- Development and training of the neural network to predict the mmWave beam

General Objectives:

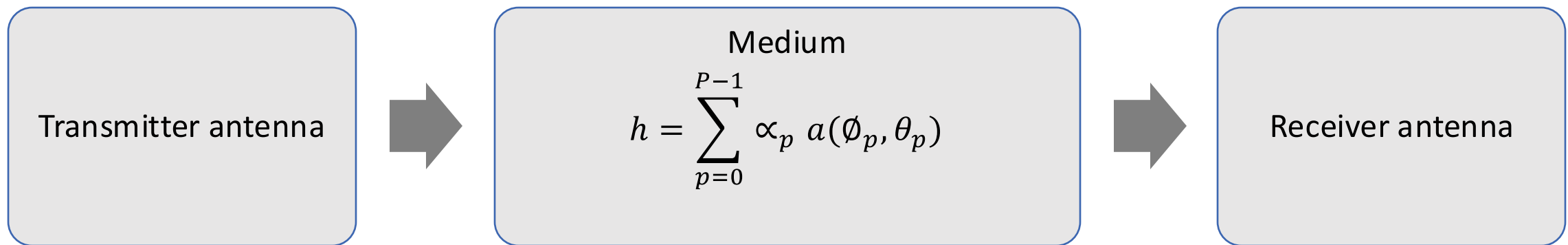
- Engage in Laboratory activities
- Understand the basis of lab research
- Apply principles learnt in other courses



1. How does a radar work?



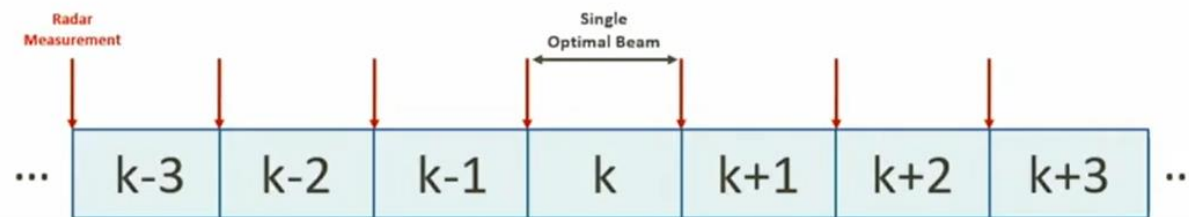
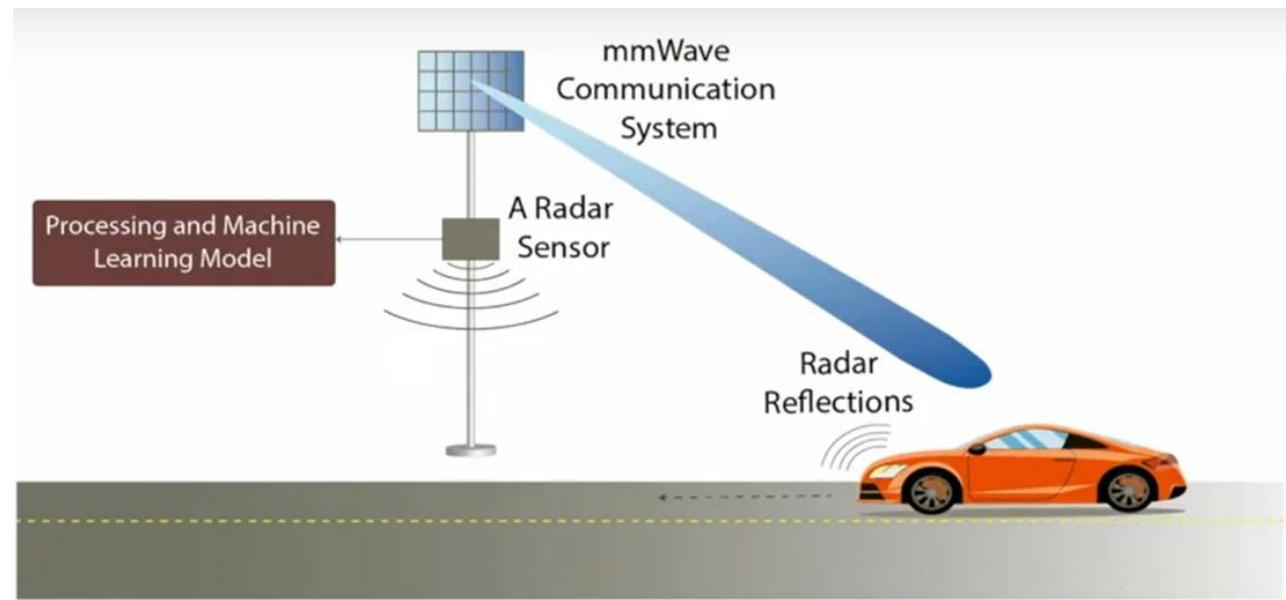
2. How does mmWave communication system work?



Adopted System Model



- Communication model
 - mmWave antenna array with large number of elements.
 - Single mobile receiver
 - Narrowband channel
- Radar model
 - FMCW radar at the base station (BS)
 - 1 TX / Mr RX antennas
- Problem and solution
 - Predict the optimal beam given radar measurements
 - Deep Learning
- Operation
 - Offline beam training
 - Online beam prediction





- Base station → FMCW (Frequency Modulated Continuous Wave) radar
- Radar objective → provide observations of environment **to the mobile unit**

How to obtain radar measurement?

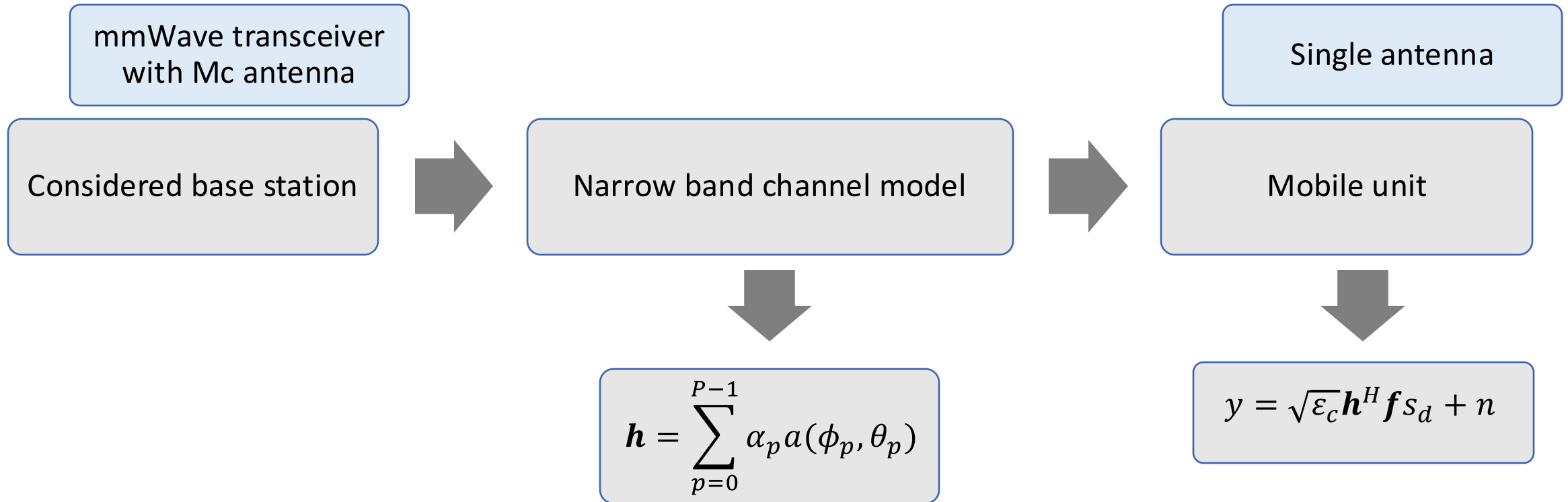
- FMCW radar transmits chirp signals → frequency changes continuously with time

$$s_{chirp}^{tx}(t) = \begin{cases} \sin\left(2\pi\left[f_c t + \frac{\mu}{s} t^2\right]\right), & \text{if } 0 \leq t \leq T_c \\ 0, & \text{otherwise} \end{cases}$$

- Single radar measurement obtained from frame of duration T_f
- In each frame, L chirp waves are transmitted with T_s waiting time between them
- Overall transmitted signal represented by:

$$s_{frame}^{tx}(t) = \sqrt{\varepsilon_t} \sum_{l=0}^{L-1} s_{chirp}^{tx}(t - (T_c + T_s) \cdot l), 0 \leq t \leq T_f$$

Communication model

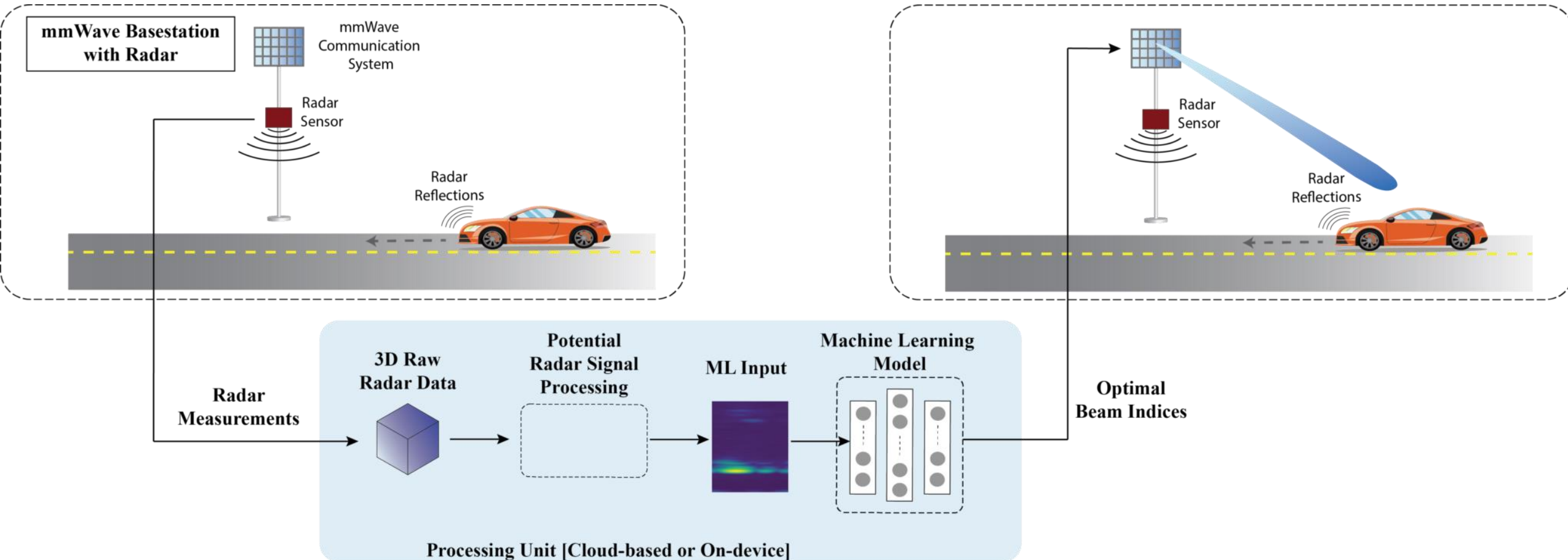


- For selecting beamforming vectors $\rightarrow$ define beamforming codebook of n vectors
- Index of optimal beam $\rightarrow$ obtained by SNR maximization problem

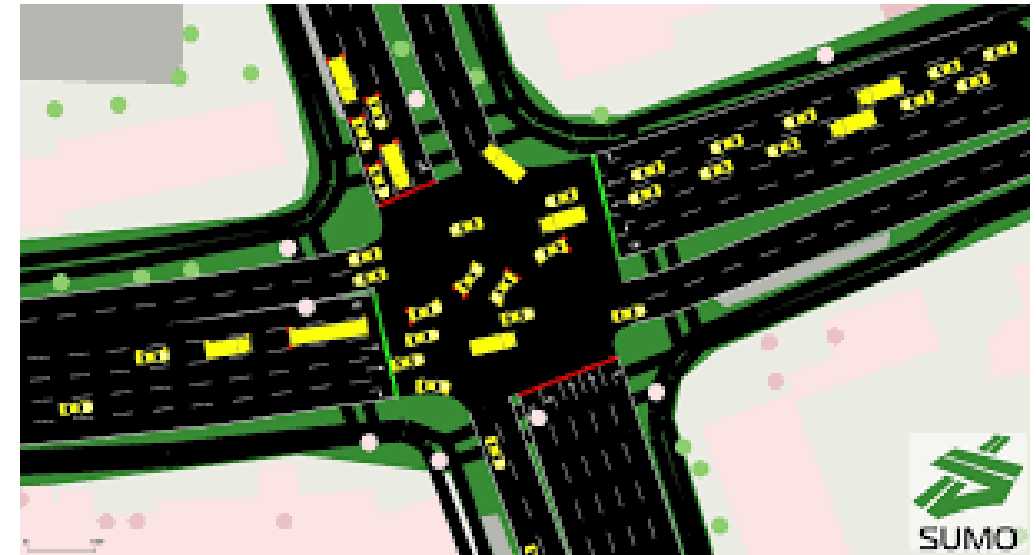
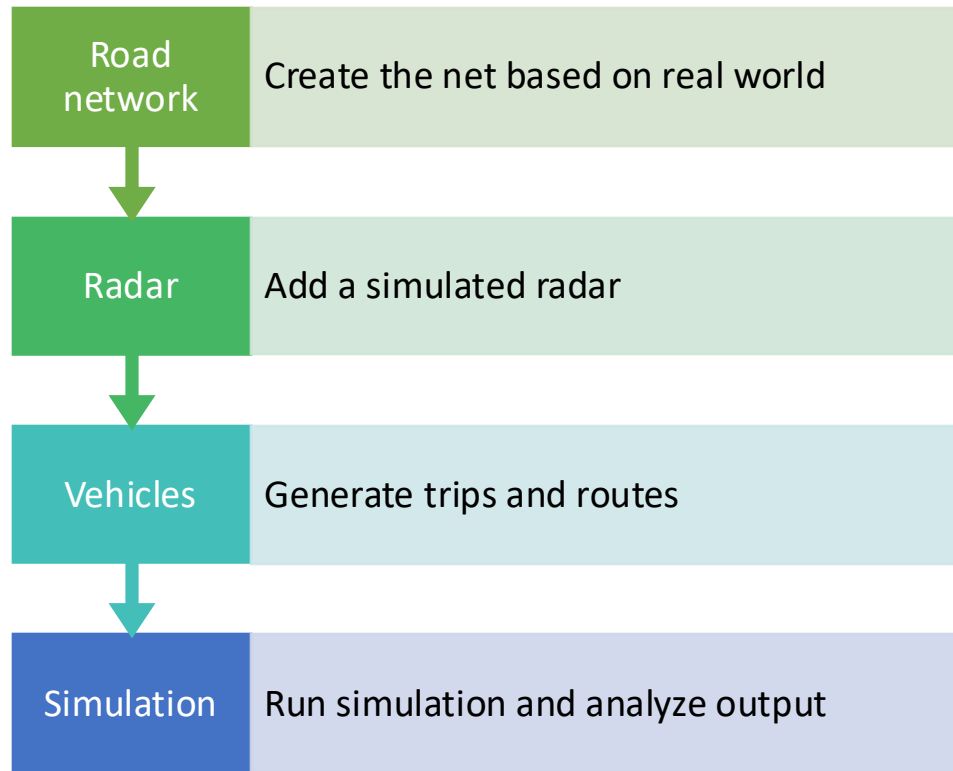
Framework



■ FMCW Radar Sensing

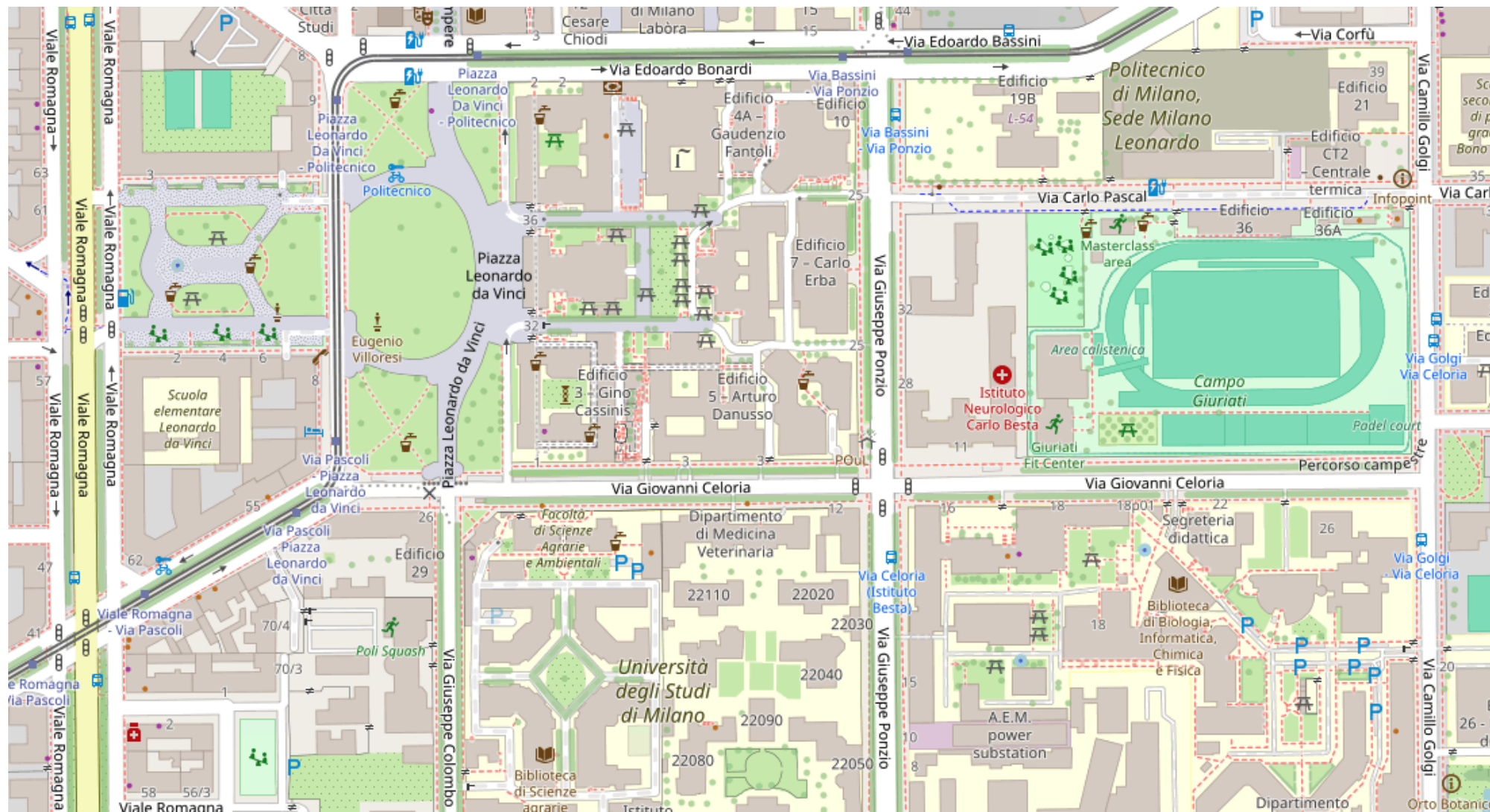


SUMO- Simulation of a real world environment

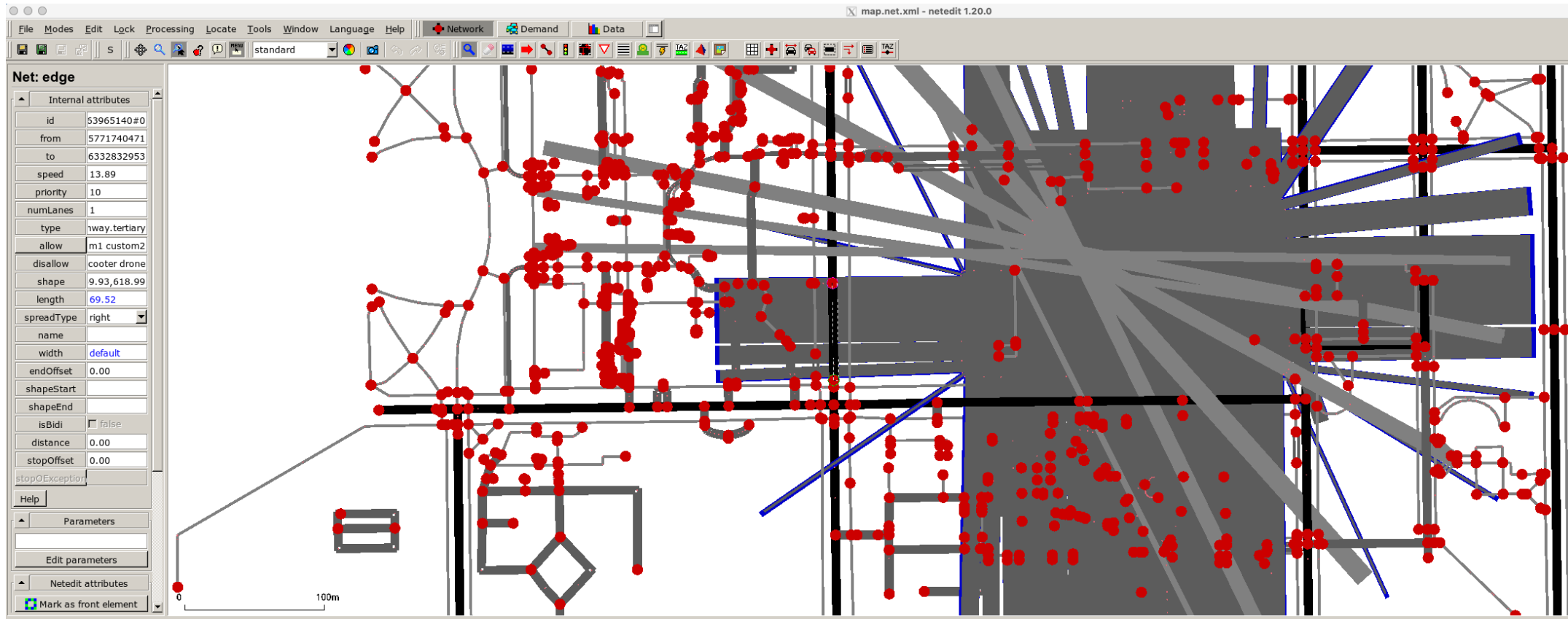




Area of the simulation (from OSM)



Creating the net (2)



● 1185 nodes

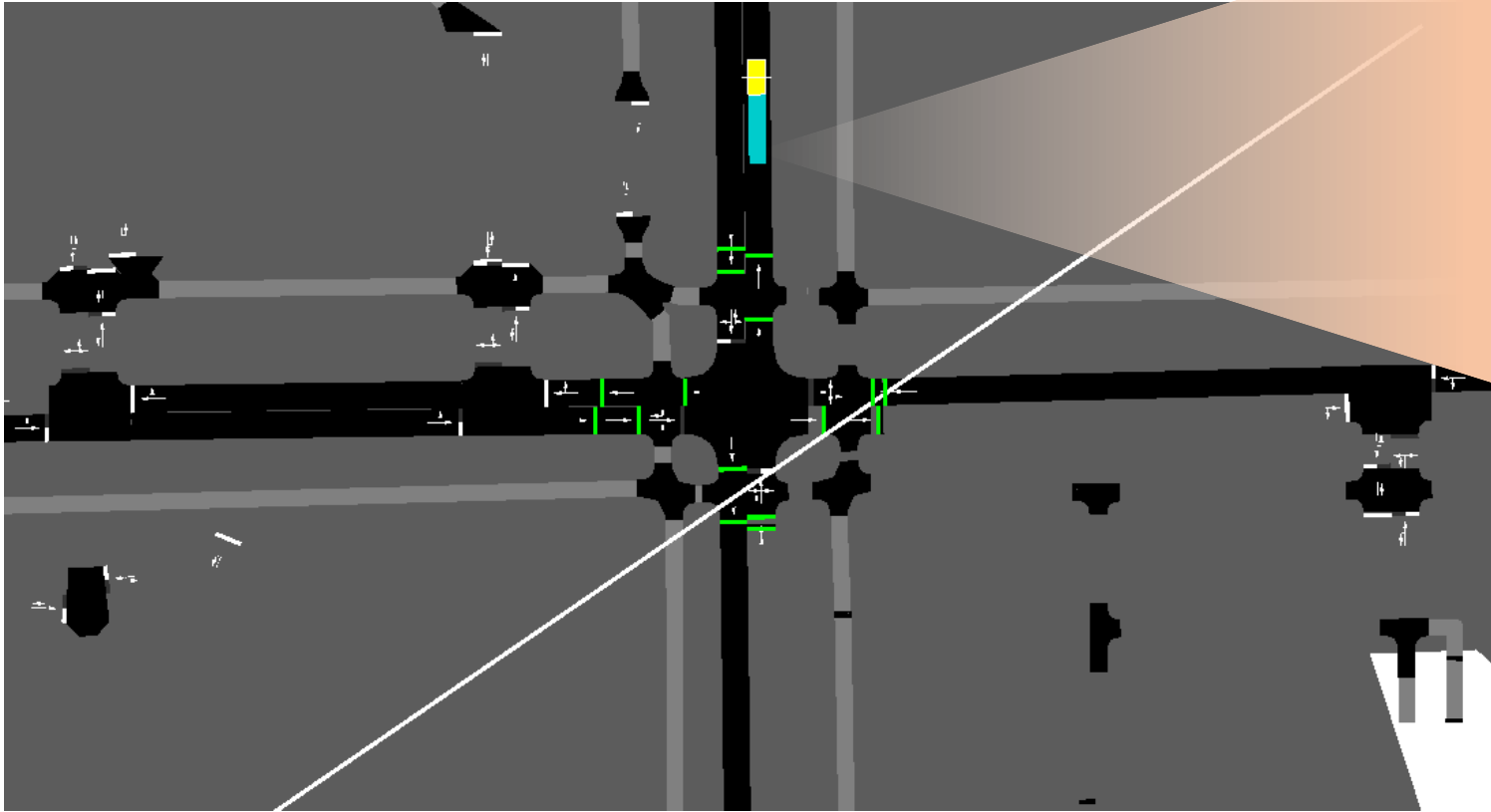
1828 edges

Parsing sumo-net from '/Users/luigilizzini/map.net.xml' ... done (170ms).
Import done:
1185 nodes loaded.

Example of attribute:

```
<type id="highway.cycleway" priority="1" numLanes="1" speed="5.56" allow="bicycle" oneway="0" width="2.00"/>
```

Adding a detector



- A detector will monitor traffic alongside the lane
- It's possible to arrange the length of sight of the detector
- A detector doesn't simulate radar measurements properly

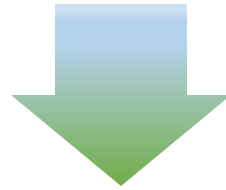
Croassroads between via Ponzio and via Celoria

Trips and routes



A script python
generates all
possible trips

A trip is
expressed in a
«from-to» form



Duarouter assigns
to each vehicle the
optimal route

A route is defined by
the starting time and
the succession of lanes
percurred





```
<detector xmlns:xsi="http://www.w3.org/2001/XMLSchema-
  interval begin="0.00" end="1000.00"
  id="radar_0"
  sampledSeconds="225.16"
  nVehEntered="34"
  nVehLeft="34"
  nVehSeen="34"
  meanSpeed="2.27"
  meanTimeLoss="5.57"
  meanOccupancy="9.41"
  maxOccupancy="74.99"
  meanMaxJamLengthInVehicles="0.17"
  meanMaxJamLengthInMeters="0.81"
  maxJamLengthInVehicles="2"
  maxJamLengthInMeters="10.00"
  jamLengthInVehiclesSum="170"
  jamLengthInMetersSum="810.27"
  meanHaltingDuration="57.67"
  maxHaltingDuration="129.00"
  haltingDurationSum="173.00"
  meanIntervalHaltingDuration="57.67"
  maxIntervalHaltingDuration="129.00"
  intervalHaltingDurationSum="173.00"
  startedHalts="4.00"
  meanVehicleNumber="0.26"
  maxVehicleNumber="2" />
```

✓ CONS

SUMO doesn't offer the opportunity to simulate a FMCW radar, so we do need a real dataset

^ PROS

The simulation can set the conditions in which the experiment could be performed

Conclusions



Simulation



Real testbed



Better Dataset



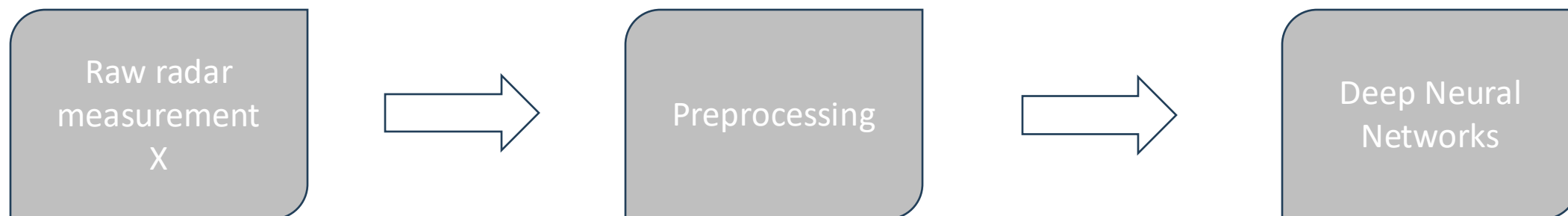
The simulation can help:

- Validate real data
- Better design the real experiment
- Fix equipment requirements



WHY:

In the previous steps, we have obtained the relevant radar information, but it is raw and unprocessed. Therefore, we chose to process the radar data with a suitable pre-processing step before feeding it into the deep neural network.





HOW:

Three different radar preprocessing approaches are proposed to extract relevant features from the raw radar measurements and prepare them for input to the Deep Neural Networks.

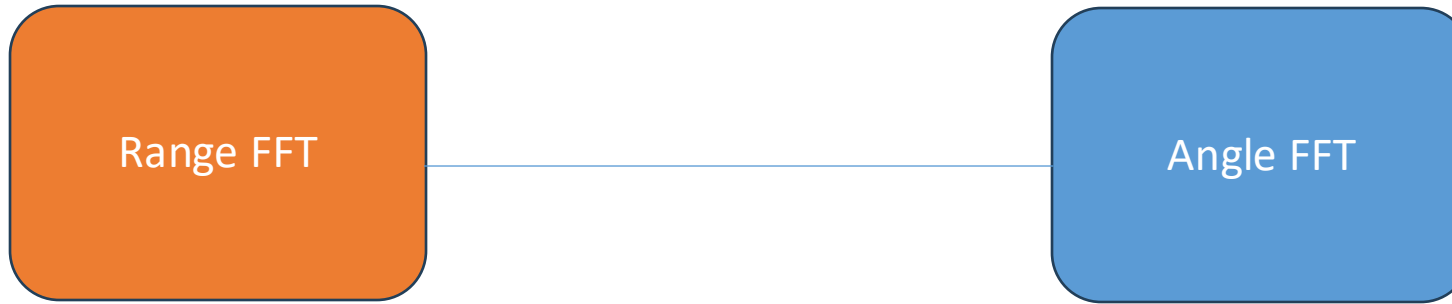
The main steps include: Range FFT; Angle FFT; Velocity FFT.

These steps result in three types of maps: **range-velocity**, **range-angle**, and a full **3D radar cube**, which includes all these features.

Range-Angle Maps



This approach aims at utilizing the range and angle information.



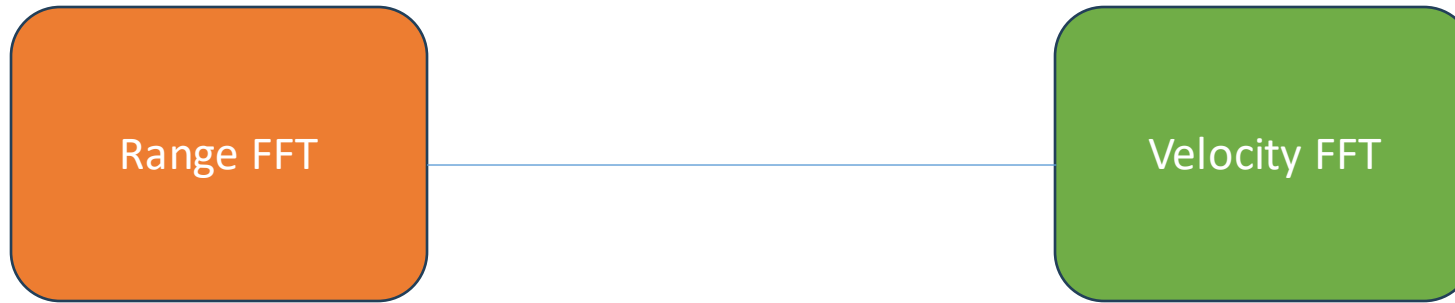
By combining resulting range-angle information for each chirp sample, we get the range-angle maps. In a simplified way, the described operation can be mathematically written as

$$\mathbf{H}_{\text{RA}} = \Psi_{\text{RA}}^{\text{P}}(\mathbf{X}) = \sum_{a=1}^A |\mathcal{F}_{2\text{D}}(\mathbf{X}_{:, :, a})|$$

Range-Velocity Maps



To construct these maps from the radar measurements, two FFTs through the time samples and chirp samples are applied.

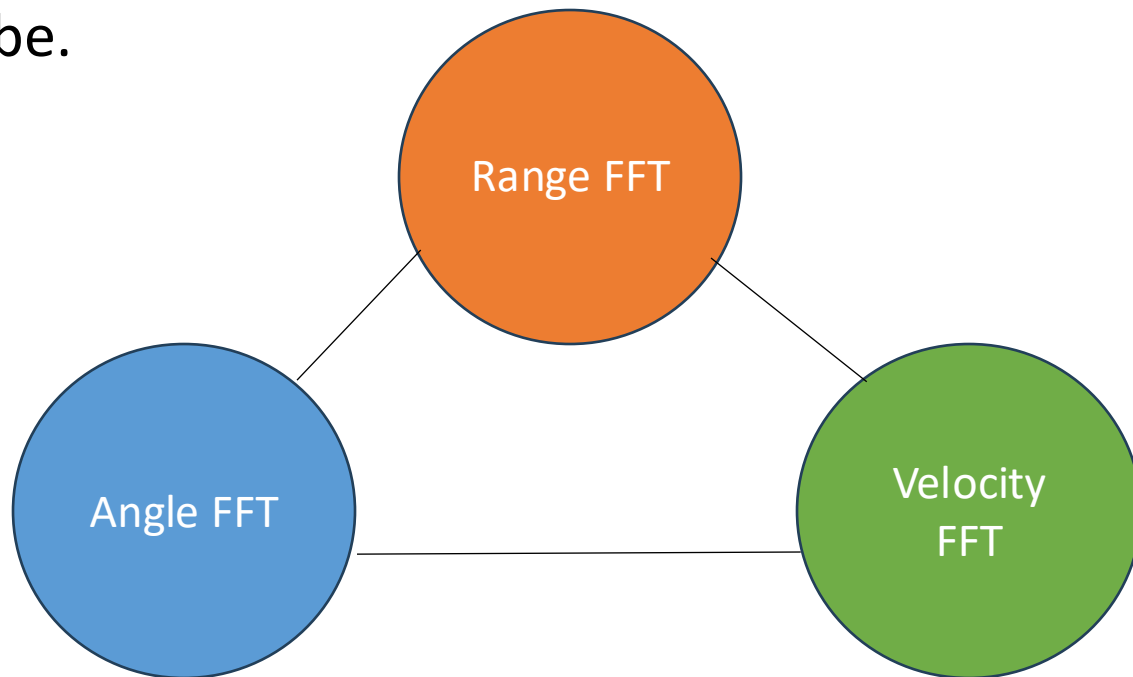


By combining the range-velocity information of the different receive antenna samples, we obtain the final range-velocity map. This operation can be written as

$$\mathbf{H}_{\text{RV}} = \Psi_{\text{RV}}^{\text{P}}(\mathbf{X}) = \sum_{m=1}^M |\mathcal{F}_{2\text{D}}(\mathbf{X}_{m,:,:})|$$



Without a dimensionality reduction, we apply the range, velocity, and angle FFTs, and obtain the radar cube.



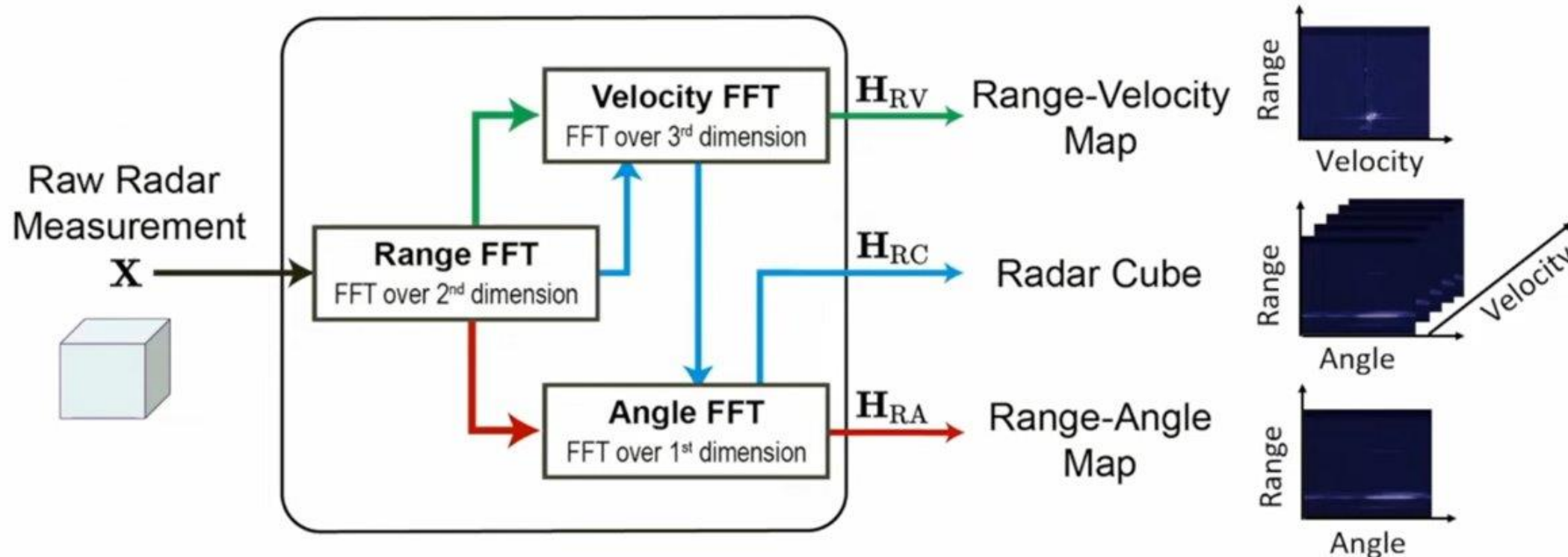
The operation can be mathematically described as

$$\mathbf{H}_{\text{RC}} = \Psi_{\text{RC}}^{\text{P}}(\mathbf{X}) = |\mathcal{F}_{3\text{D}}(\mathbf{X})|.$$

Framework



- Extract range, angle, and velocity features (Maps computation)
- Lower dimensional interpretable data for further processing



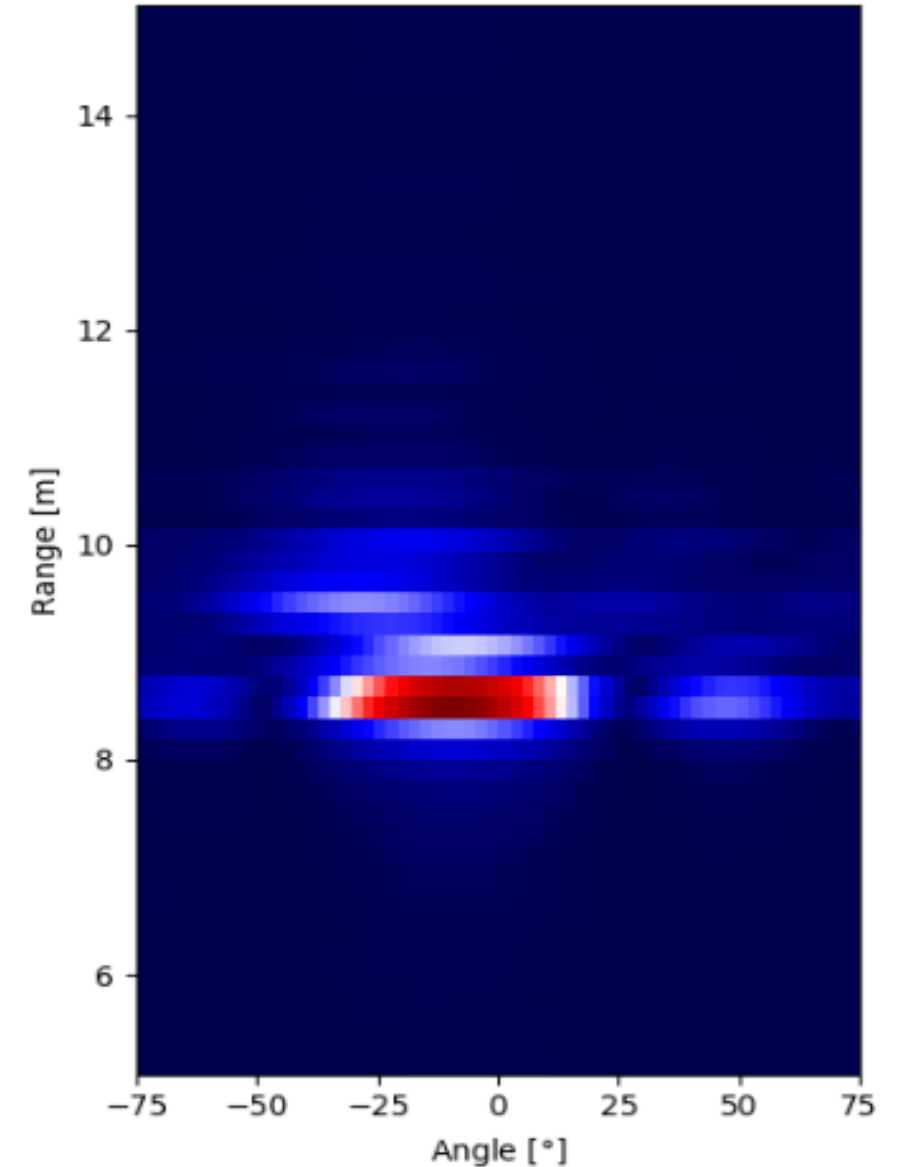
Radar Data Visualizations



- Range-angle map

sample_idx = 10

The Range-Angle plot shows a concentration of energy around a range of 8 meters and angle between -25° and $+25^\circ$. The car is on the left part of the camera image, moving towards to the radar device.



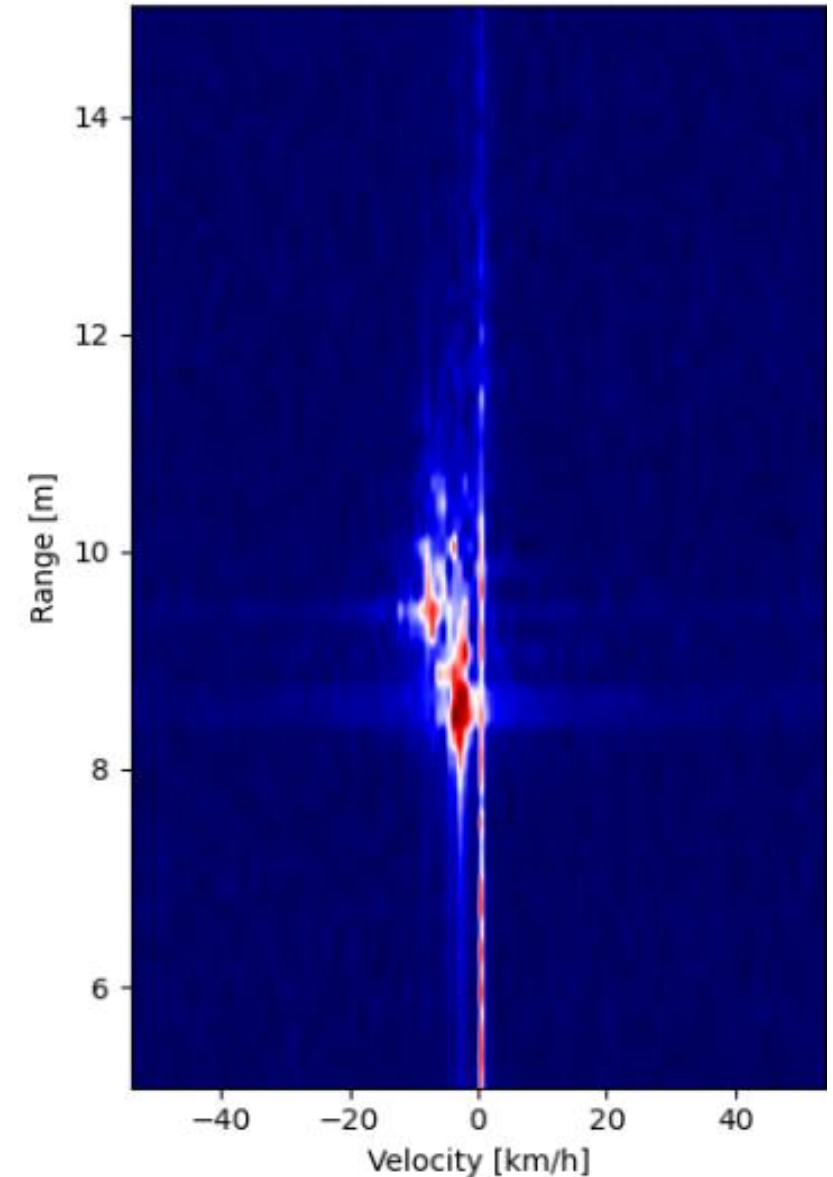
Radar Data Visualizations



■ Range-Velocity Map

sample_idx = 10

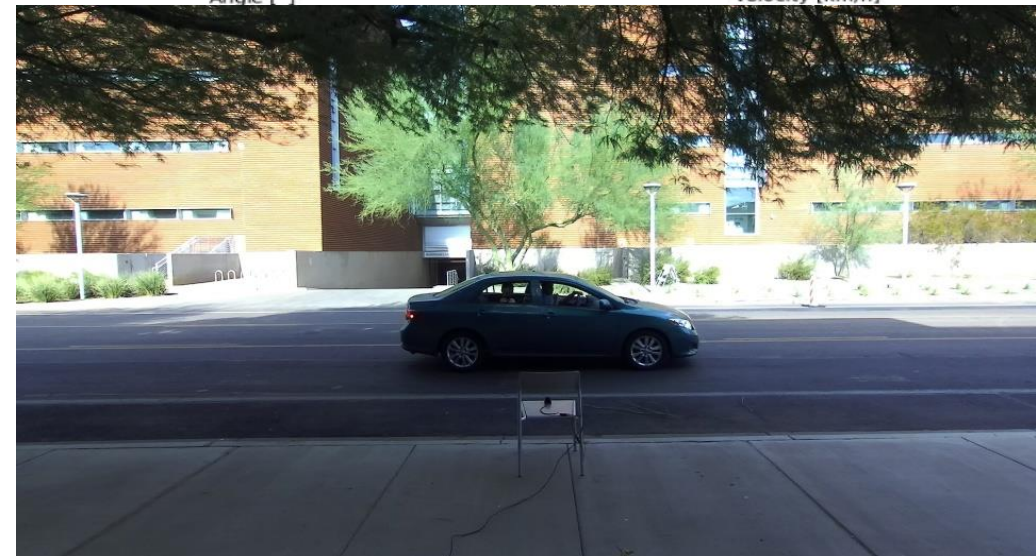
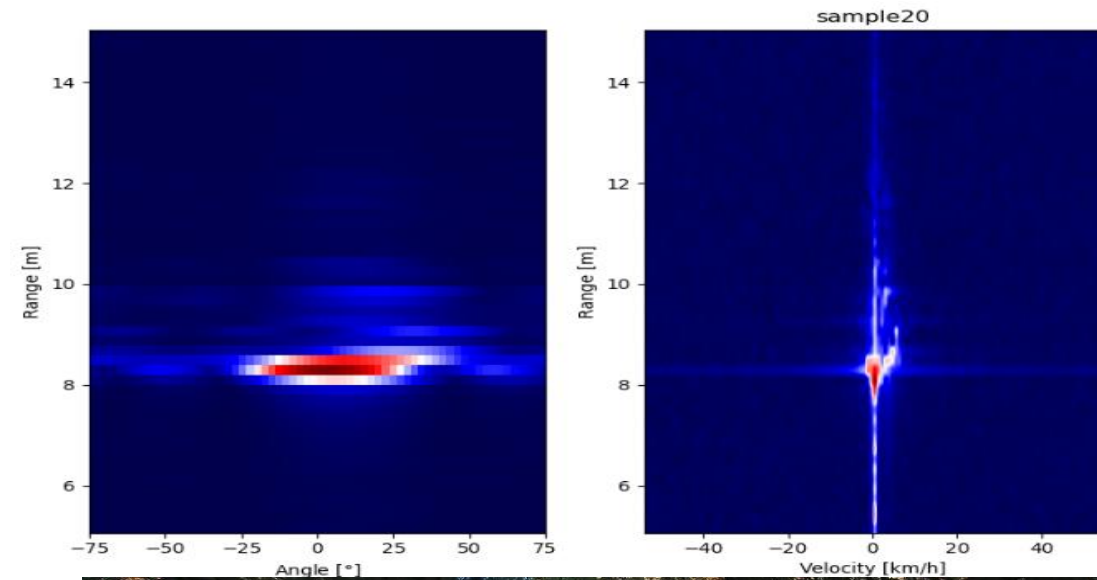
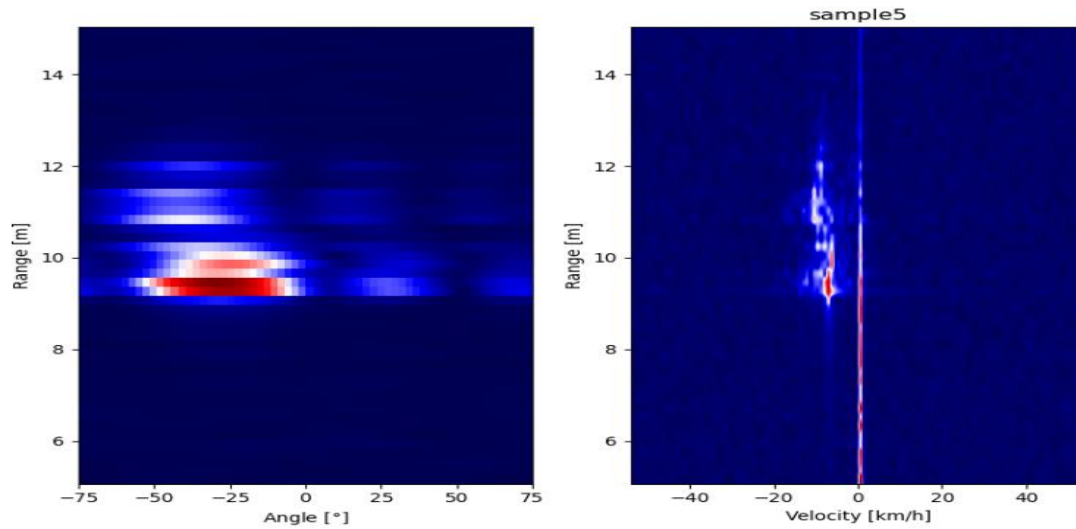
The range in this plot is focused between 8 meters and 10 meters, with the brightest concentration around 8 meters. The velocity is around 0 km/h, suggesting that the car in the scene is nearly stationary or at a very low speed.



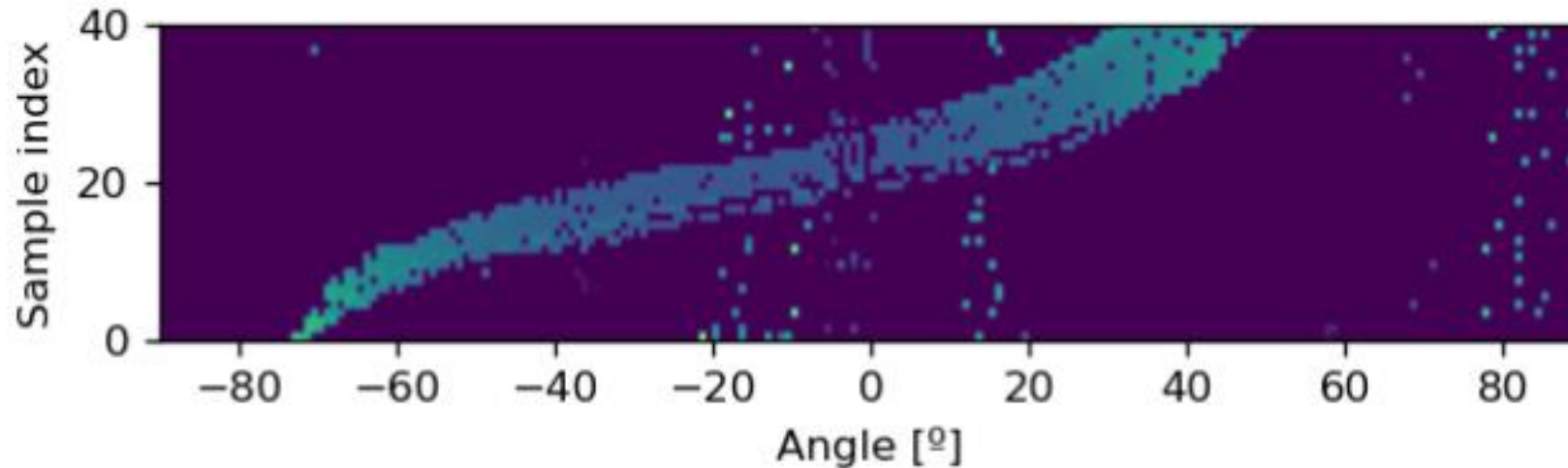
Radar Data Visualizations



- Different samples (sample 5 and 20)



LIDAR data Visualization



The plot shows the first 40 samples scanned by the Lidar sensor. This LIDAR visualization can help track the vehicle's movement more accurately, thus providing richer environmental information that helps to improve the accuracy of beam prediction. However, LIDAR data typically contains more information than radar data, resulting in larger data sizes and potentially longer processing times.

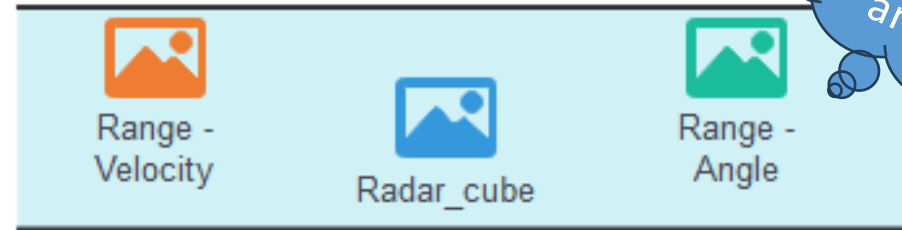
Conclusion of Radar Preprocessing part



In conclusion, radar preprocessing plays a crucial role in the proposed radar-aided beam prediction framework by extracting relevant features from the raw radar data and preparing them for input to the Deep Neural Networks. This enables the DNNs to learn the complex mapping between the radar measurements and the optimal beamforming vector, leading to improved beam prediction accuracy.

Deep Learning Part

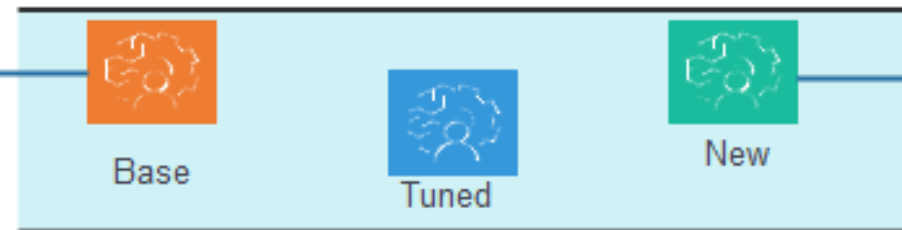
Input



Range - Angle 4 and 64

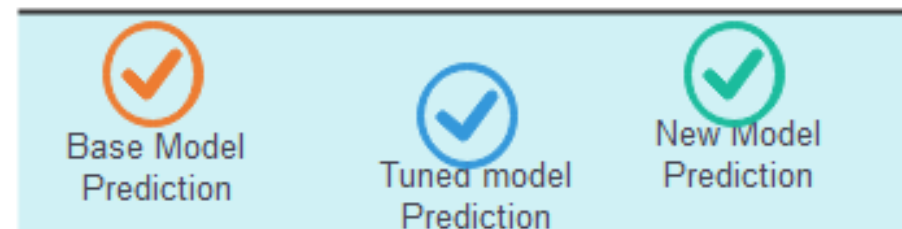
Models

5 Conv Nets + 4 Avg Pooling layers + 3 Fully Connected layers



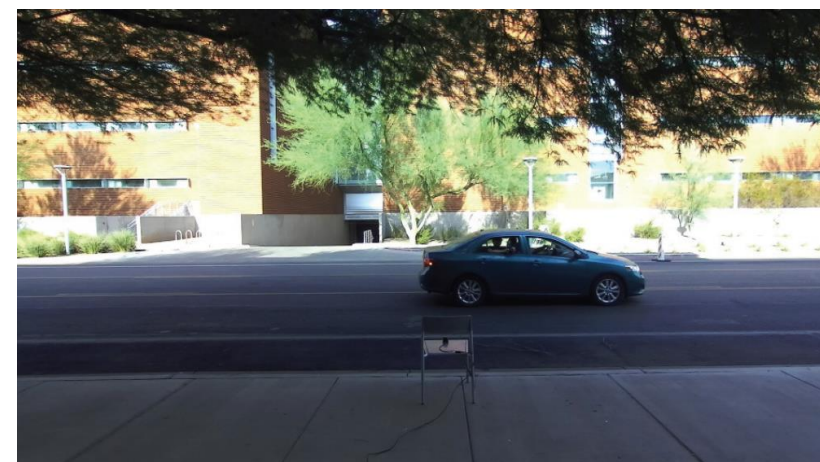
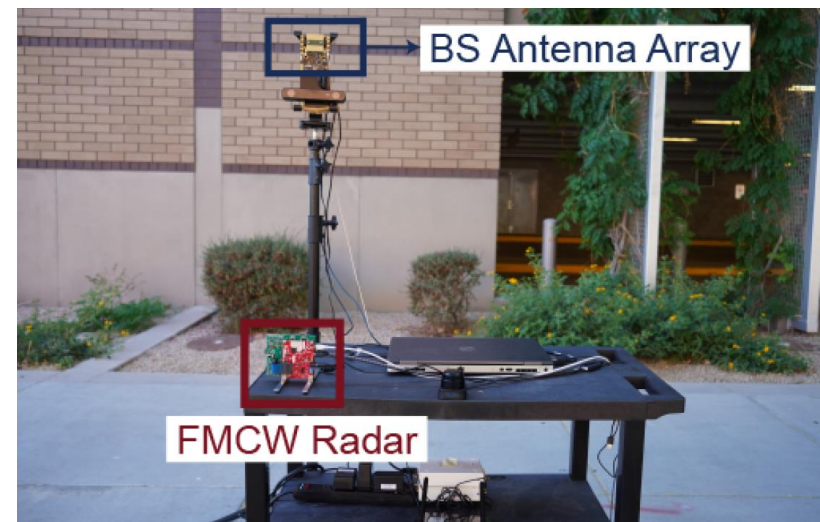
9 Conv Nets + 2 Avg Pooling + 2 Max Pooling layers + 4 Fully Connected layers using Dropouts and residual links

Output



Input dataset has been created with this setup:

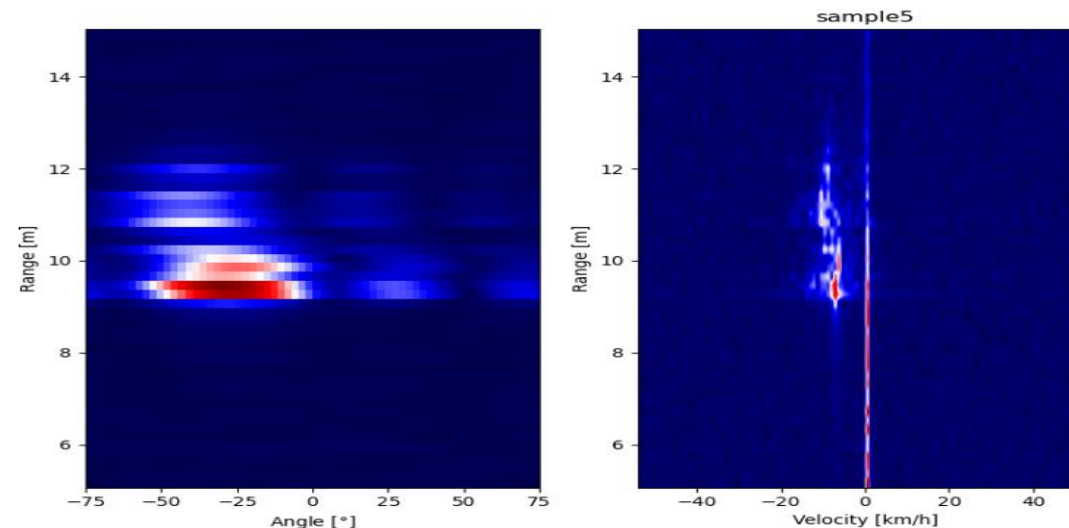
- Transmitter (Mobile User):
 - 60 GHz omnidirectional transmitter
- Receiver (Base Station):
 - 60 GHz ULA with 16 antennas
 - 64 beams (oversampled)
 - Off-the-shelf FMCW radar with 4 RX antennas
- Radar parameters
 - 750 MHz bandwidth
 - 256 samples/chirps – 128 chirps/frame
 - 45m max. range – 56 km/h max velocity



■ Inputs:

- **Training_dataset:** 4199 images (70%)
- **Validation_dataset:** 1172 images (20%)
- **Testing_dataset:** 593 images (10%)
- **Size of images:** (4,256,128), (1,256,128), (1,256,64), (1,256,4) for Radar-Cube, Range-Velocity, Range-Angle 64 and Range-Angle 4, respectively.

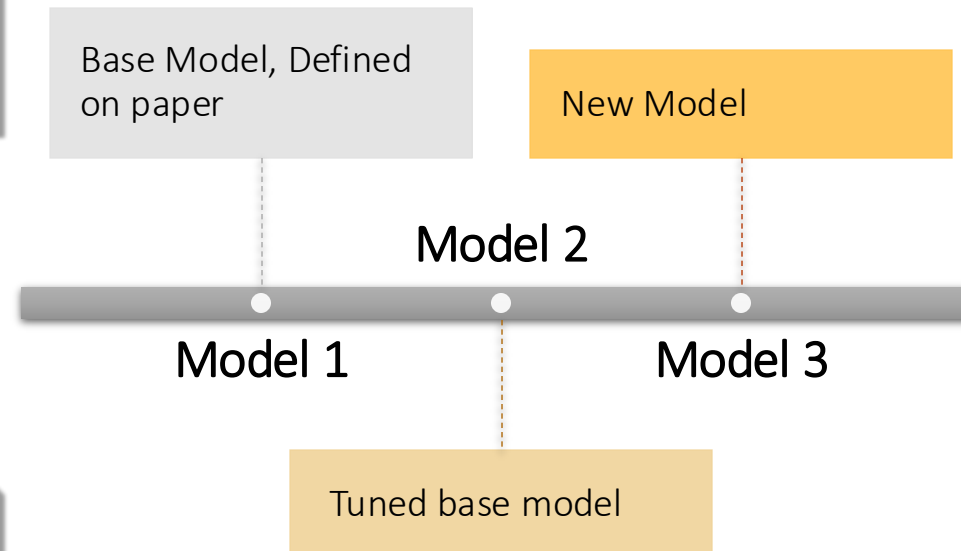
As previously presented, some examples of Range-Angle (left) and Range-Velocity (right) maps:





**5 Conv Nets + 4 Avg Pooling
layers +
3 Fully Connected layers**

**After tuning,
Epochs = 40, batch_size = 8,
lr = 1e-3**



**9 Conv Nets + 2 Avg Pooling+2
Max Pooling layers +
4 Fully Connected layers using
Dropouts and residual links**



■ Model Summaries

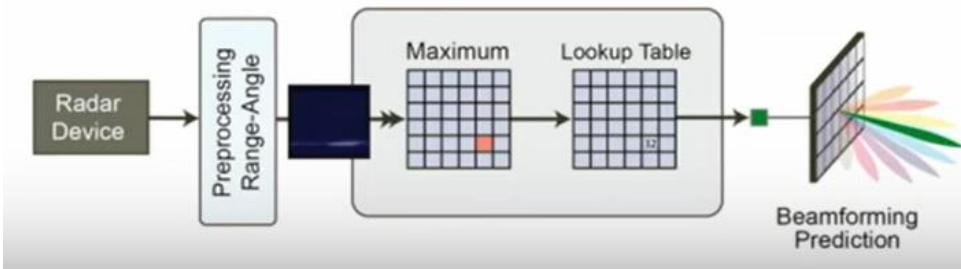
Base Model			New Model		
Layer (type)	Output Shape	Param #	Layer (type)	Output Shape	Param #
Conv2d-1	[-1, 8, 256, 128]	296	Conv2d-1	[-1, 8, 256, 128]	296
Conv2d-2	[-1, 16, 256, 128]	1,168	Conv2d-2	[-1, 16, 256, 128]	1,168
AvgPool2d-3	[-1, 16, 128, 128]	0	Conv2d-3	[-1, 8, 256, 128]	1,160
Conv2d-4	[-1, 8, 128, 128]	1,160	Conv2d-4	[-1, 4, 256, 128]	292
AvgPool2d-5	[-1, 8, 64, 64]	0	AvgPool2d-5	[-1, 4, 128, 64]	0
Conv2d-6	[-1, 4, 64, 64]	292	Conv2d-6	[-1, 8, 128, 64]	296
AvgPool2d-7	[-1, 4, 32, 32]	0	Conv2d-7	[-1, 16, 128, 64]	1,168
Conv2d-8	[-1, 2, 32, 32]	74	Conv2d-8	[-1, 8, 128, 64]	1,160
AvgPool2d-9	[-1, 2, 16, 16]	0	Conv2d-9	[-1, 4, 128, 64]	292
Linear-10	[-1, 256]	131,328	AvgPool2d-10	[-1, 4, 64, 64]	0
Linear-11	[-1, 128]	32,896	Conv2d-11	[-1, 2, 64, 64]	74
Linear-12	[-1, 64]	8,256	MaxPool2d-12	[-1, 2, 32, 32]	0
Total params: 175,470			Linear-13	[-1, 1024]	2,098,176
Trainable params: 175,470			Dropout-14	[-1, 1024]	0
Non-trainable params: 0			Linear-15	[-1, 512]	524,800
Input size (MB): 0.50			Dropout-16	[-1, 512]	0
Forward/backward pass size (MB): 9.43			Linear-17	[-1, 256]	131,328
Params size (MB): 0.67			Dropout-18	[-1, 256]	0
Estimated Total Size (MB): 10.60			Linear-19	[-1, 64]	16,448
			Total params: 2,776,658		
			Trainable params: 2,776,658		
			Non-trainable params: 0		
			Input size (MB): 0.50		
			Forward/backward pass size (MB): 11.73		
			Params size (MB): 10.59		
			Estimated Total Size (MB): 22.82		

Results

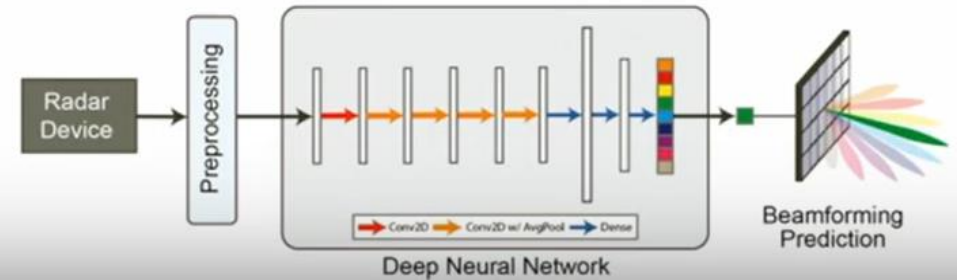


Results obtained from Base Model

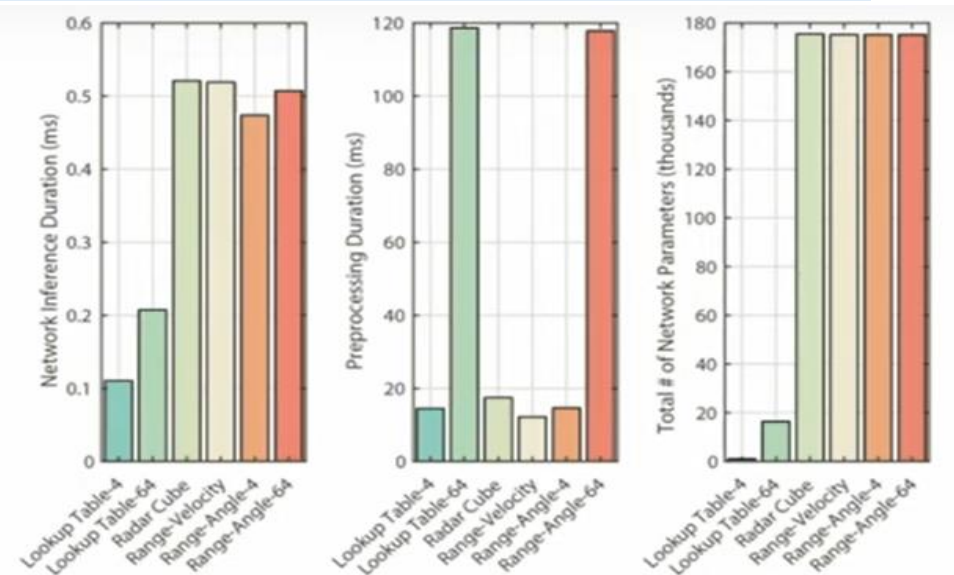
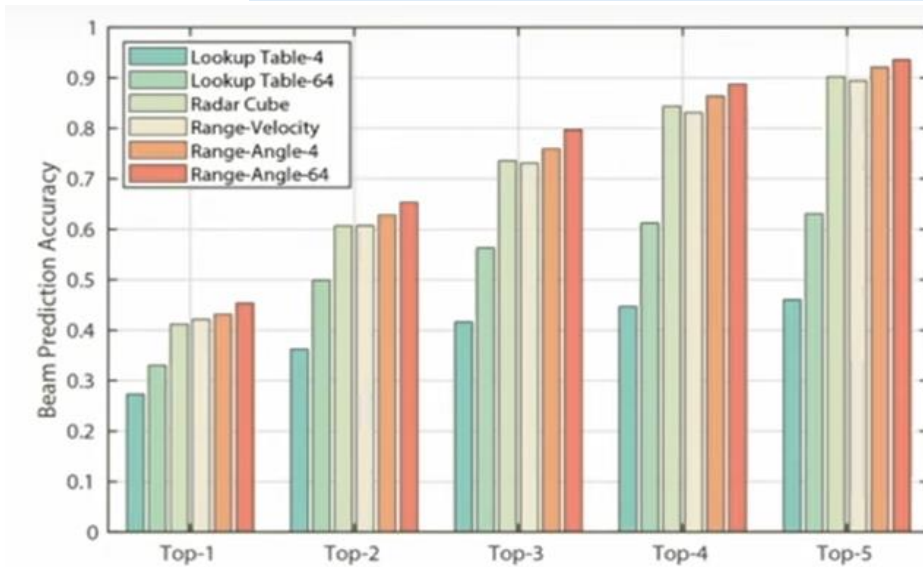
Prediction By Traditional method



Prediction By DL method

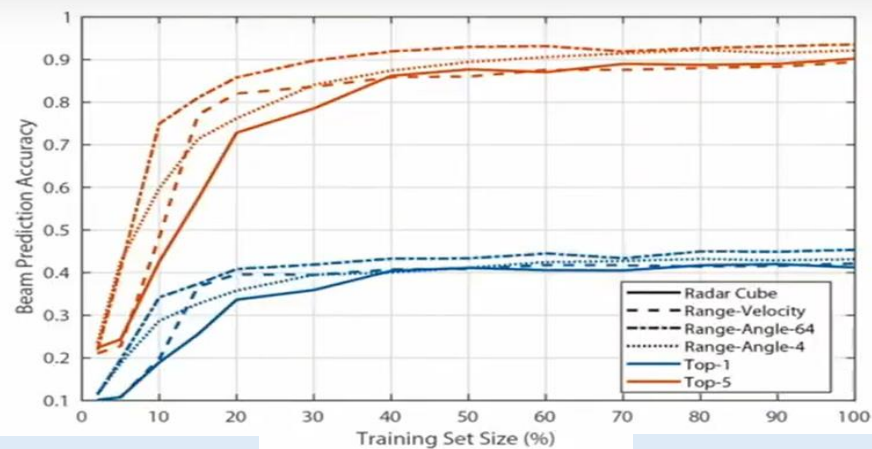


Accuracy of Base Model using different inputs of Traditional way and DL

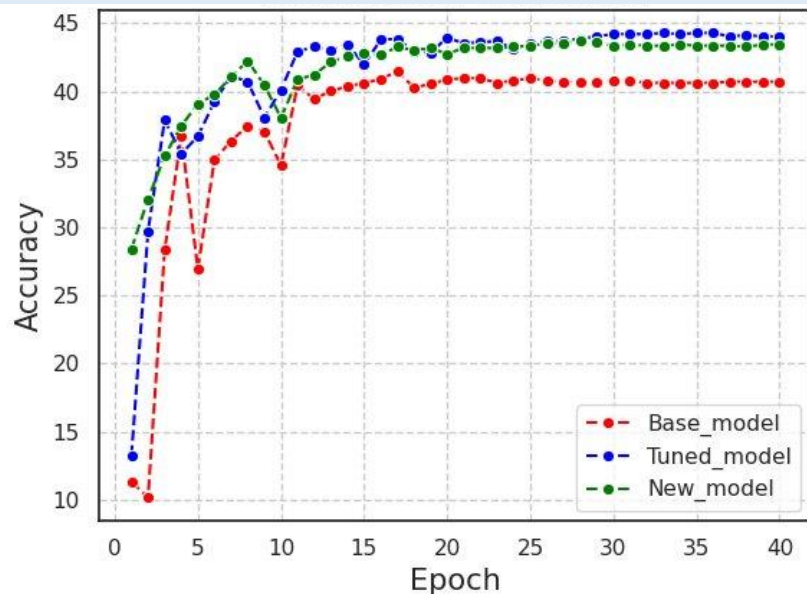




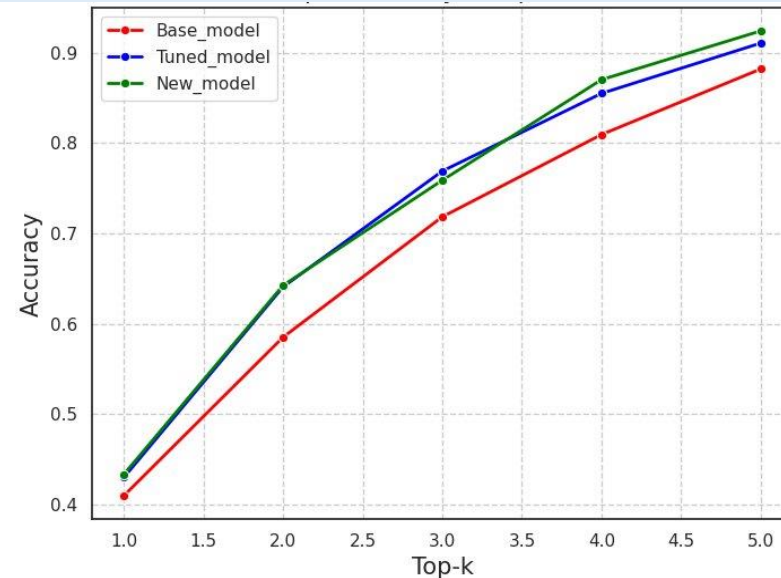
Effect of training dataset on Base Model



Test Accuracy comparison of three models using radar cube input



Top-K Accuracy comparison of three models using radar cube input



Conclusions



- With the help of DL Models, we got around 90% Accuracy in top-5
- Although DL Model showed better results, they need more resources to run. So trading off is crucial.
- The tuned model and new one both showed 5% better accuracy.
- Traditional methods can be a good choice when we have some concern about resources, as they have less parameters.



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THANK YOU
